

## Step 5: Data Visualization and Analysis

Data visualization and analysis are essential steps in the Credit Card Approval Prediction System. This phase focuses on exploring the dataset to understand its structure, identify patterns, detect relationships among variables, and gain insights that support the development of an accurate machine learning model. Various statistical methods and visualization techniques are used to analyze the applicant data before preprocessing and model training.

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### Step 5.1: Understanding the Dataset

The first step is to understand the structure and contents of the dataset. The dataset is loaded into a Pandas DataFrame, and its dimensions, column names, and data types are examined. This helps identify the available features and understand how they contribute to the prediction process.

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### Step 5.2: Displaying the First Few Records

The first few rows of the dataset are displayed to verify that the data has been loaded correctly. This provides a quick overview of the applicant information, including demographic, financial, and credit-related attributes.

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### Step 5.3: Examining Dataset Information

The overall information of the dataset is analyzed to determine the number of records, number of features, data types, and memory usage. This step helps identify numerical and categorical variables that require different preprocessing techniques.

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### Step 5.4: Generating Statistical Summary

A statistical summary of the numerical features is generated to understand the distribution of the data. Measures such as count, mean, standard deviation, minimum value, maximum value, and quartiles are examined to identify unusual values and understand the characteristics of each feature.

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### Step 5.5: Identifying Missing Values

The dataset is analyzed to detect missing or null values in each column. Identifying missing data is important because incomplete information may reduce the accuracy of the machine learning model. Appropriate preprocessing techniques are planned based on the missing value analysis.

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## **Step 5.6: Visualizing the Distribution of Features**

Different visualization techniques are used to understand how individual features are distributed. Histograms and count plots help identify the frequency of numerical and categorical values, while box plots assist in detecting outliers that may affect model performance.

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## **Step 5.7: Analyzing Relationships Between Features**

Relationships among different variables are examined using correlation analysis and scatter plots. A correlation matrix and heatmap are generated to identify strongly related features. Understanding these relationships helps in feature selection and improves model performance.

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## **Step 5.8: Performing Univariate Analysis**

Univariate analysis is carried out by examining one feature at a time. This analysis helps understand the distribution, variability, and characteristics of each variable independently. It also assists in identifying skewed distributions and unusual observations.

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## **Step 5.9: Performing Bivariate Analysis**

Bivariate analysis is performed to study the relationship between two variables. This analysis helps determine how different applicant attributes influence the credit card approval outcome and identifies features that are strongly associated with the target variable.

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## **Step 5.10: Detecting Outliers**

Outliers are identified using graphical methods such as box plots. Extreme values may negatively affect machine learning models; therefore, they are carefully analyzed before deciding whether to remove or transform them during preprocessing.

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## **Step 5.11: Interpreting the Analysis Results**

The insights obtained from statistical analysis and data visualization are summarized to understand the overall characteristics of the dataset. The identified patterns, relationships, and potential issues provide valuable guidance for data preprocessing, feature engineering, and machine learning model development.

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## **Step 5.12: Preparing for Data Preprocessing**

Based on the observations obtained during visualization and analysis, the dataset is prepared for preprocessing. Missing values, outliers, inconsistent data, and irrelevant features identified during this phase are addressed in the next stage to improve the quality of the dataset before training the machine learning model.